

SOLUTIONS

1. $\frac{7}{8}$ of 1008 - $\frac{3}{4}$ of 568 = $\left(1008 \times \frac{7}{8}\right) - \left(568 \times \frac{3}{4}\right)$
 $= 882 - 426 = 456.$
2. Let the number be x .
 Then, $x - \frac{3}{5}x = 50 \Leftrightarrow \frac{2}{5}x = 50$
 $\Leftrightarrow x = \left(\frac{50 \times 5}{2}\right) = 125.$
3. Let the number be x .
 Then, $x + \frac{2}{5}x = 455 \Leftrightarrow \frac{7}{5}x = 455 \Leftrightarrow x = \left(\frac{455 \times 5}{7}\right) = 325.$
4. Let the number be x . Then, $x \times \frac{2}{3}x = 864$
 $\Leftrightarrow \frac{2}{3}x^2 = 864$
 $\Leftrightarrow x^2 = \left(\frac{864 \times 3}{2}\right) = 1296 \Leftrightarrow x = \sqrt{1296} = 36.$
5. Let the number be x . Then, $\frac{x-4}{6} = 8$
 $\Leftrightarrow x - 4 = 48$
 $\Leftrightarrow x = 52.$
 $\therefore \frac{x-2}{5} = \frac{52-2}{5} = \frac{50}{5} = 10.$
6. Let the number be x .
 Then, $13x = x + 180 \Leftrightarrow 12x = 180$
 $\Leftrightarrow x = \frac{180}{12} = 15.$
7. Let the number be x .
 Then, $2x + 3 \times 42 = 238 \Leftrightarrow 2x + 126 = 238$
 $\Leftrightarrow 2x = 112 \Leftrightarrow x = 56.$
 $\therefore$ Required sum = $3x + 2 \times 42 = 3 \times 56 + 2 \times 42$
 $= 168 + 84 = 252.$
8. Let the number be x . Then, $\frac{1}{3}$ of $\frac{1}{4}$ of $x = 15$
 $\Leftrightarrow x = 15 \times 12 = 180.$
 So, required number = $\left(\frac{3}{10} \times 180\right) = 54.$
9. Let the number be x .
 Then, $\frac{3}{5}$ of $\frac{2}{3}$ of $x - \frac{2}{5}$ of $\frac{1}{4}$ of $x = 288$
 $\Leftrightarrow \left(x \times \frac{3}{5} \times \frac{2}{3}\right) - \left(x \times \frac{2}{5} \times \frac{1}{4}\right) = 288$
 $\Leftrightarrow \frac{2}{5}x - \frac{1}{10}x = 288$
 $\Leftrightarrow \frac{3x}{10} = 288 \Leftrightarrow x = \left(\frac{288 \times 10}{3}\right) = 960.$
10. Let the number be x .
 Then, $3(2x + 9) = 75 \Leftrightarrow 2x + 9 = 25$
 $\Leftrightarrow 2x = 16 \Leftrightarrow x = 8.$
11. Let the number be x .
 Then, $\frac{3}{4}x - \frac{1}{3}x = 60$
 $\Leftrightarrow \frac{5x}{12} = 60$
 $\Leftrightarrow x = \left(\frac{60 \times 12}{5}\right) = 144.$
12. Let the number be x .
 Then, $\frac{x}{5} + 4 = \frac{x}{4} - 10 \Leftrightarrow \frac{x}{4} - \frac{x}{5} = 14$
 $\Leftrightarrow \frac{x}{20} = 14 \Leftrightarrow x = 14 \times 20 = 280.$
13. Let the number be x .
 Then, $x - 24 = \frac{4}{7}x \Leftrightarrow x - \frac{4}{7}x = 24$
 $\Leftrightarrow \frac{3}{7}x = 24 \Leftrightarrow x = \left(\frac{24 \times 7}{3}\right) = 56.$
 $\therefore$ Sum of the digits = $(5 + 6) = 11.$
14. Let the number be x .
 Then, $15x - x = 196$
 $\Leftrightarrow 14x = 196$
 $\Leftrightarrow x = 14.$
15. Let the number be x .
 Then, $\frac{x}{4} = x - 21 \Leftrightarrow x = 4x - 84$
 $\Leftrightarrow 3x = 84 \Leftrightarrow x = 28.$
16. Let the number be x .
 Then, $\left(\frac{1}{5}x + 4\right) = \left(\frac{1}{4}x - 10\right)$
 $\Leftrightarrow \frac{x}{20} = 14 \Leftrightarrow x = 14 \times 20 = 280.$
17. Let the number be x .
 Then, $x - 12 = 20\% \text{ of } x \Leftrightarrow x - \frac{x}{5} = 12$
 $\Leftrightarrow \frac{4x}{5} = 12 \Leftrightarrow x = \left(\frac{12 \times 5}{4}\right) = 15.$
18. Let the number be x .
 Then, $\frac{1}{7}x - \frac{1}{11}x = 100 \Leftrightarrow \frac{4x}{77}$
 $= 100 \Leftrightarrow x = \frac{7700}{4} = 1925.$
19. Let the number be x .
 Then, $\left(\frac{1}{2}x + \frac{1}{5}x\right) - \frac{1}{3}x = \frac{22}{3} \Leftrightarrow \frac{11x}{30}$
 $= \frac{22}{3} \Leftrightarrow x = \left(\frac{22 \times 30}{3 \times 11}\right) = 20.$
20. Let the number be x .
 Then, $2x + 20 = 8x - 4 \Leftrightarrow 6x = 24$
 $\Leftrightarrow x = 4.$